

LESSON PLAN

Name of the Programme:	B.Tech(Hons.)	
Title of the Course:	Programming in C	
Course code	CS1003	
Semester:	I	
Names of the Course Instructor(s)	Dr Dattatreya Mankame, Dr. J. Cynthia, Dr. Mahadev Kalyanrao Patil, Prof. Vinod K Raju, Prof. Neha Najem and Prof. Manasa H C	
Course credit	No. of hours per week (3+0+2)	Total no. of Teaching hours
04	05 Hours	75 Hours

Session	Module	Topic	Pedagogy/Activities	Prereading/Reference
1	M1	Syllabus and Lesson plan explanation, Introduction to C and Computer fundamentals	PPT presentation and Whiteboard	Let Us C Ch. 1 and 12, C Programming Absolute Beginner's Guide (CABG) Ch.1,2
2		Structure of C Program, Memory Layout of C Programs, Phases of developing a computer program in C		
3		Program execution stages Preprocessor, Compiler, Linker, Loader, Library Functions, Various components of a computer.		
4		Lab program 1: Program to add numbers using macros	Hands-on coding session	
5	M2	Constants, Variables, Rules for Constructing Variables, understanding different Data Types - range, size, and appropriate usage, C Keywords	PPT presentation and Whiteboard	Let Us C Ch.1, CABG Ch.4
6		Format specifiers, Uses and Common Errors associated with them	PPT presentation and Whiteboard: Write code using different format specifiers	
7		Pre- processor Directive, Macros	PPT presentation and Whiteboard	Let Us C Ch.12, CABG Ch.7
8		Lab program 2: Program to swap two numbers with and without using a third variable	Hands-on coding session	Let Us C Ch. 2 and 4
9		Operators – Arithmetic operators, Hierarchy of operations, Precedence, Relational Operators	PPT presentation and Whiteboard: Write code using different Operators	
10		Logical Operators, Conditional Operators, Bitwise Operators.		
11		Assignment Operators, Type conversion and Type		

		casting		
12		Lab program 3: Program to create a menu-driven basic arithmetic calculator	Hands-on coding session	
13	M3	Decision Control: if Statement, else-if Statement, Ladder if Statement	PPT presentation and Whiteboard: - Flowcharts to visualize logic - Code tracing exercises	Let Us C Ch. 3-5
14		Nested if Statement, Switch Statement		
15		Loop – for, while Statement		
16		Lab program 4: Program to print a Fibonacci series and compute GCD of two numbers	Hands-on coding session	
17	M3	Do while Statement, Difference between all looping statements	PPT presentation and Whiteboard: - Flowcharts to visualize logic - Code tracing exercises	Let Us C Ch. 5,6,8
18		Break, continue, exit statements		
19		Introduction to Functions: Function Declaration and Definition		
20		CIE1 Exam	Assessment	
21		Function Prototype, Calling a Function, Return Value	PPT presentation and Whiteboard: Write code to understand the concepts of functions & scope of variables	Let Us C Ch. 8
22		Parameters and Arguments: Local, Global, and Static Variables		
23		Arguments scope of variables, Pass by value		
24		Lab program 5: Program to print a Fibonacci series and compute GCD of two numbers using functions	Hands-on coding session	
25		Pass by reference	PPT presentation and Whiteboard: - Flowcharts to visualize logic - Code tracing exercises	Let Us C Ch. 8-10
26		Understanding the use of pointers, Basic concept of recursion		
27		Recursion problems – Factorial, GCD		
28		Lab program 6: Program to print a Fibonacci series and compute GCD of two numbers using recursion	Hands-on coding session	
29		Recursion problems – Fibonacci series	PPT presentation and Whiteboard: - Flowcharts to visualize logic - Code tracing exercises	Let Us C Ch. 9-10

		Difference between iteration, recursion and use of pointers	PPT presentation and Whiteboard	
31	M4	Introduction to arrays, initialization, indexing array elements in memory		Let Us C Ch. 13-14, CABG Ch.25
32		Lab program 7: Program to print a Fibonacci series and compute GCD of two numbers using recursion	Hands-on coding session	
33		Bounds checking, 2- dimensional, multi-dimensional arrays	PPT presentation and Whiteboard:	Let Us C Ch. 13-14, CABG Ch.21
34		Passing array elements to functions	Write code to understand how arrays are passed as arguments.	
35		Passing entire array to functions		
36		Lab program 8: Program to implement different string operations using functions	Hands-on coding session	
37	M4	Basic array operations - traversal, insertion, deletion, updation in 1D array	PPT presentation and Whiteboard:	Let Us C Ch. 13-14
38		Basic array operations - traversal, insertion, deletion, updation in 2D array	- Flowcharts to visualize logic - Code tracing exercises	
39		Array manipulation- reversing, concatenation		
40		CIE2 Exam	Assessment	
41		Array manipulation- splitting, sorting	PPT presentation and Whiteboard:	Let Us C Ch. 13-14, CABG Ch.23
42		Array Library functions	- Flowcharts to visualize logic - Code tracing exercises	CABG Ch.23
43		Strings as array of characters, strings with pointers		Let Us C Ch.15, CABG Ch.19
44		Lab program 9: Program using structures to read, write and compute average-marks of n students	Hands-on coding session	
45		String Operations	PPT presentation and Whiteboard	Let Us C Ch.15, CABG Ch.19
46		String Library Functions		
47	M5	Introduction to structure and union		Let Us C Ch.17
48		Lab program 10: Program to demonstrate different file operations	Hands-on coding session	
49		Structure definition and declaration	PPT presentation and Whiteboard:	Let Us C Ch.17, CABG Ch.27
50		Accessing structure elements	Write code to understand the concept of structure	
51		How structure elements are stored		
52		Lab practice session		

53		Array of structures	PPT presentation and Whiteboard	Let Us C Ch.17, 19, CABG Ch.27
54		Passing structure as parameter		
55		File Operations: Creating new file		
56		Lab Internal CIE3 component Exam	Assessment	
57		Opening and closing an existing/new file	PPT presentation and Whiteboard	Let Us C Ch.17, CABG Ch.28
58		Reading from and writing to a file		
59		Moving to a specific location in a file		
60		CIE3 Exam	Assessment	

Assessment and Evaluation

Continuous Internal Evaluation (CIE) :70%	
Components of CIE	Weightage of each component
CIE-1 - Completion of self-learning modules Component 1 - Online assessment Component-2	5 marks out of 70 15 marks out of 70
CIE-2 - Theory	25 marks out of 70
CIE-3 - Completion of self-learning modules Component 1 - Continuous Lab assessment + Viva Component 2 - Internal lab practical exam Component 3	5 marks out of 70 5 marks out of 70 15 marks out of 70

Semester End Exams (SEE): 30%	
Mode of Exam:	Theory Exam
Weightage of exam	30%

Course Outcomes- Programme Outcomes Matrix

Course Outcomes		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
a.	Understand and analyse the concepts of the number system, computer memory, and the fundamental concepts of the C programming language.	3	3	2	2	2			3		1		1
b.	Demonstrate proficiency in using fundamental data types, operators, and console input/output functions in C	3	3	2	2	2			3		1		1
c.	Design and implement C programs using decision-making constructs, loops, arrays, strings, functions, and pointers to solve mathematical and logical problems.	3	3	2	2	2			3		1		1
d.	Analyse and evaluate the programming concepts based on user-defined data types and file handling in C language.	3	3	2	2	2			3		1		1

Note: High: 3, Medium: 2, Low: 1

Rubrics for evaluation of CIE (separate for each CIE component)

Total Marks		20
CIE-1 Components		Rubrics for Assessment
CIE-1	Component 1	Self-learning module completion for 5 marks. Specified modules in the CodeTantra must be completed within the given timeframe. Faculty will verify module completion through the CodeTantra faculty interface (<i>Refer Table-1</i>)
	Component 2	Test will be conducted for 15 marks and will be evaluated according to the Scheme of Evaluation prepared by the team of course faculty.

Table-1:

CIE 1 and CIE 3 Graded Component - 1 (5 Marks):

Assess students' ability to complete assigned self-learning modules on the CodeTantra platform within the given timeframe.

Criteria	Fully Completed (5 Marks)	Mostly Completed (3-4 Marks)	Partially Completed (1-2 Marks)	Incomplete (0 Marks)
Completion of self-learning modules	Successfully completes all modules on the CodeTantra platform before the deadline.	Completes more than 50% of the modules.	Completes less than 50% of the modules.	Fails to complete any module within the given time.

Total Marks		25
CIE-2 Components		Rubrics for Assessment
CIE-2	Theory	Test will be conducted for 25 marks and will be evaluated according to the Scheme of Evaluation prepared by the team of course faculty.

Total Marks		25
CIE-3 Components		Rubrics for Assessment
CIE-3	Component 1	Self-learning module completion for 5 marks. Specified modules in the CodeTantra must be completed within the given timeframe. Faculty will verify module completion through the CodeTantra faculty interface (Refer Table-1)
	Component 2	Continuous Lab assessment + Viva for 5 marks (Refer Table-2) (10 lab programs 10 marks each, 100 marks reduced to 5 marks)
	Component 3	Internal lab practical exam for 15 marks (Refer Table-2) (3 lab programs 10 marks each, 30 marks reduced to 15 marks)

Table-2:

CIE 3 Graded Component - 2 (5 Marks), Component - 3 (15 Marks)

Assess students' performance through continuous lab work and their understanding via viva.

Lab assessment + Viva (10 Marks per program)				
Criteria	Measuring Methods	Excellent (2 Marks)	Good (1 Mark)	Poor (0 Marks)
Understanding of problem statement (2 Marks)	Observations	The student exhibits a thorough understanding of the requirements for the problem.	The student has a sufficient understanding of the requirements for the problem.	The student does not have a clear understanding of the requirements for the problem.
Execution (2 Marks)	Observations	The Student demonstrates the execution of the program with optimized code and shows the output with appropriate validations and test cases handled.	The Student demonstrates the execution of the program without optimization of the code and shows the output.	The Student has not executed the program.
Results and Documentation	Observations	Documentation with appropriate comments	Documentation with few comments and	Documentation with no comments

(2Marks)		and output cases are noted in data sheets.	few output cases are noted in data sheets.	and no output cases are noted in data sheets.
Viva (4 Marks)	Observations	The student is eager to answer all questions and provides meaningful answers to the questions asked (4– 3 M)	The student tries to answer the questions partially and answers with some cues and help. (2-1 M)	The student does not answer the questions even with cues and help. (0 M)